

**ADOPTION AND USE OF ARTIFICIAL INTELLIGENCE FOR JOURNALISM
PRACTICE AMONG JOURNALISTS IN THE FEDERAL CAPITAL TERRITORY,
ABUJA, NIGERIA**

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Abstract

This study examined the adoption and use of Artificial Intelligence (AI) in journalism practice among journalists in the Federal Capital Territory (FCT), Abuja, Nigeria, focusing on the gap between high awareness and limited operational use. The study addressed how contextual constraints shape the integration of AI within newsroom practices in a developing media environment. A quantitative survey design was adopted. Data were collected from 170 journalists across print, broadcast, and digital media organisations using a structured questionnaire, yielding 164 valid responses. The study was guided by Technological Determinism and Diffusion of Innovation theory. Data were analysed using descriptive and inferential statistics to assess awareness, usage patterns, perceived impact, and adoption barriers. The findings showed that awareness of AI was widespread among journalists, yet its application remained limited and task-specific. AI was primarily used for editing, transcription, and content generation, with minimal integration into core editorial processes. Respondents acknowledged its role in improving efficiency and supporting investigative work; however, this did not translate into sustained or comprehensive use. Structural barriers, including unreliable electricity supply, high internet access costs, inadequate infrastructure, and limited technical skills, constrained adoption. Ethical concerns relating to data accuracy, editorial control, and content originality also influenced usage decisions. The results indicated a clear disconnect between awareness and practical utilisation, shaped by institutional and environmental conditions. The study concluded that AI adoption in journalism within the FCT is incremental and conditioned by structural realities rather than technological availability alone. It recommended institutional frameworks for AI integration, targeted training for journalists, curriculum reform in media education, improved digital infrastructure, and the development of ethical standards to guide responsible AI use.

Keywords: Artificial Intelligence, Journalism Practice, Technology Adoption, Media Innovation, Nigeria, Newsroom Integration

Introduction

The integration of Artificial Intelligence (AI) into journalism represents one of the most consequential shifts in the evolution of modern media practice. While early forms of newsroom automation emerged through rule-based systems designed to generate routine reports, particularly in financial and sports journalism, the field has since advanced towards sophisticated algorithmic and data-driven models capable of performing complex editorial functions (Graefe, 2016; Diakopoulos, 2019). In recent years, the proliferation of machine learning and generative AI technologies has further accelerated this transformation, enabling news organisations to automate content production, enhance investigative reporting through large-scale data analysis, and personalise news delivery at unprecedented scale (Beckett, 2019; Thurman, Dörr, & Kunert, 2017). These developments have given rise to what is now widely conceptualised as computational journalism, in which human editorial processes are increasingly interwoven with algorithmic decision-making systems (Carlson, 2015; Lewis, Guzman, & Schmidt, 2019).

In advanced media systems, particularly in North America and Europe, AI has moved beyond experimental deployment to become an embedded component of newsroom operations. Major organisations such as Reuters, the Associated Press, and the BBC have adopted AI-driven tools for automated reporting, real-time data processing, audience analytics, and misinformation detection, which enhance both the speed and scale of news production (Beckett, 2019; Diakopoulos, 2019). Similar patterns are observable in parts of Asia and Europe, where AI is increasingly used to support editorial decision-making and audience engagement strategies. In these contexts, scholars caution that AI adoption is not merely a technical process but is shaped by institutional priorities, economic considerations, and professional norms (Napoli, 2019).

Beyond these highly resourced environments, the trajectory of AI adoption becomes more uneven and contextually contingent. In transitional and developing media systems, including parts of Southern Europe, Latin America, and Africa, AI integration is often partial, experimental, or constrained by structural limitations (Canavilhas, 2022; Noain-Sánchez, 2022). Within the African context, emerging evidence suggests that while awareness of AI technologies is growing, their practical application remains limited due to infrastructural deficits, high operational costs, limited technical expertise, and weak institutional support systems (Ogola, 2023; Mutsvairo, Bebawi, & Borges-Rey, 2020). These conditions reflect broader patterns of digital inequality, where disparities in access to technological resources and capabilities shape the uneven distribution of innovation across global media landscapes.

Despite increasing scholarly attention to AI in journalism, much of the existing literature is characterised by two significant limitations. First, it tends to adopt an implicitly deterministic perspective, assuming that technological innovation will inevitably translate into widespread adoption and transformation of journalistic practice. Second, empirical research remains disproportionately concentrated in Western contexts, with relatively limited attention to how AI is experienced, interpreted, and used within developing media environments. Even within African scholarship, studies have often focused on awareness levels or general perceptions, with insufficient attention to the structural and institutional factors that mediate actual use in specific contexts.

These limitations are particularly evident in Nigeria, where the media landscape is undergoing rapid digital expansion alongside persistent infrastructural and economic constraints. As one of Africa's largest and most dynamic media markets, Nigeria presents a critical site for examining the intersection between technological innovation and structural limitation. Within the Federal Capital Territory (FCT) of Abuja, home to major broadcast, print, and digital media organisations, journalists operate in a complex environment characterised by increasing exposure to digital tools, yet constrained by irregular electricity supply, high data costs, and uneven access to advanced technologies. While anecdotal and preliminary evidence suggests growing familiarity with AI tools among journalists, their sustained integration into newsroom workflows remains uncertain and uneven.

This apparent disjuncture between awareness and utilisation raises a fundamental analytical question that existing studies have yet to address adequately: why does increased exposure to AI technologies not translate into consistent and effective adoption in journalism practice in resource-constrained environments? Addressing this question is essential not only for understanding the dynamics of technological change in the Nigerian media sector but also for challenging broader assumptions about the universality of digital transformation in journalism. Theoretically, this study interrogates this problem through a critical engagement with Technological Determinism, which posits that technological innovation acts as the primary driver of social and institutional change (McLuhan, 1964). While this perspective has been influential in explaining the transformative potential of media technologies, it has been widely criticised for underestimating the role of socio-economic structures, human agency, and contextual constraints in shaping technological outcomes (Winner, 1980). To address these limitations, the present study draws on complementary insights from the Diffusion of Innovation theory (Rogers, 2003) and the Technology Acceptance Model (Davis, 1989), both of which emphasise that technology adoption is contingent upon factors such as perceived usefulness, accessibility, cost, and environmental conditions. This integrated theoretical approach allows for a more nuanced understanding of AI adoption as a socially mediated process rather than a technologically determined outcome.

Against this backdrop, this study examines the adoption and use of Artificial Intelligence in journalism practice among journalists in the Federal Capital Territory, Abuja, Nigeria. It investigates levels of awareness, patterns of utilisation, types of AI tools employed, and the structural, professional, and ethical factors influencing their adoption. By situating empirical findings within a broader theoretical and contextual framework, the study moves beyond descriptive accounts of technological presence to provide a deeper explanation of the conditions under which AI is, or is not, integrated into journalistic practice.

This study is significant for three key reasons. First, it contributes to the growing body of literature on AI and journalism by providing context-specific empirical evidence from a Global South perspective and addressing a critical gap in existing research. Second, it advances theoretical debates by demonstrating that technological change in journalism cannot be adequately explained through deterministic frameworks alone but must be understood in relation to infrastructural, economic, and institutional realities. Third, it offers practical insights for policymakers, media organisations, and journalism educators seeking to develop context-responsive strategies for integrating AI into journalism in developing environments. In doing

so, the study provides a more grounded and practicable understanding of how AI can be meaningfully adopted within the Nigerian media landscape.

Clarification of Concepts

Artificial Intelligence (AI):

Artificial Intelligence (AI) is a branch of computer science that focuses on creating intelligent systems capable of performing tasks that usually require human reasoning or understanding. In simple terms, it seeks to make machines “think” and “act” intelligently. According to Nahal, as cited in Okiyi and Nsude (2019), AI is expected to be a significant driving force of technological advancement in the coming decade. This prediction aligns with current global trends, in which the creative and industrial sectors are seeing increasing automation driven by AI-based systems. Guanah, Obi and Ginikachukwu (2020), citing Ladi Ojora, describe AI as a process that enables computer systems to learn, analyse, and make decisions independently of human input. This perspective highlights AI's ability to mimic human cognitive abilities such as reasoning, problem-solving, language understanding, and object recognition (Nnamdi & Nwanyanwu, 2021). Essentially, AI represents a system that not only performs tasks automatically but also adapts and improves over time. Scholars such as Dhiman (2023) emphasise that AI can transform daily life and work patterns by enhancing decision-making, automating repetitive tasks, and improving the overall user experience.

Laskowski and Tucci (2024) describe AI as the replication of human cognitive processes in machines, particularly computer systems. Copeland (2024) supports this view, noting that AI enables computers to perform activities once exclusive to intelligent beings. Lateef (2024) further expands this understanding by explaining that AI involves both the scientific and engineering processes of developing intelligent systems. He categorises AI into four types: *Reactive Machines*, *Limited Memory*, *Theory of Mind*, and *Self-Aware Machines*. Reactive Machines function based on present data without learning from the past, while Limited Memory systems make decisions using historical information. The Theory of Mind category aims to develop emotionally intelligent systems that can understand human beliefs and intentions. Finally, Self-Aware Machines represent the most advanced stage, in which systems possess consciousness and self-awareness similar to those often imagined in science fiction.

Journalism Practice

Journalism involves the collection, analysis, writing, and distribution of news and information through various media platforms, including radio, television, newspapers, magazines, online platforms, and billboards. Its primary purpose is to inform the public and address their diverse information needs. Hasan (2014) explains that journalism practice revolves around answering fundamental questions, such as *Who? What? When? Where? Why?* and *How?* This process ensures that journalists provide accurate, balanced, and relevant information to the public.

Journalism encompasses every activity related to gathering, verifying, processing, and communicating news and current affairs for public consumption. As IGI Global (2024) notes, journalism includes preparing stories for dissemination across multiple media channels, whether print, broadcast, or digital. It therefore plays a critical role in ensuring that societies remain informed, engaged, and connected through credible reporting on events, trends, and issues affecting individuals and communities.

A Global Empirical Review on the Application and Challenges of AI for Journalism Practice

Across the world, researchers have shown increasing interest in how artificial intelligence is being applied in journalism and the challenges that accompany its use. For instance, Guannah (2021) conducted a study titled “*Mainstream Media and Artificial Intelligence Awareness amongst Residents of Asaba Metropolis, Delta State, Nigeria.*” The aim was to understand the extent to which traditional media, such as newspapers, radio, and television, inform and educate residents about AI applications. The study also sought to determine residents' level of AI knowledge and assess whether media messages have been effective in promoting public awareness. Guided by the Diffusion of Innovation Theory, the researcher employed a survey and in-depth interview design to gather data from residents. Findings revealed that public awareness and understanding of AI among Asaba residents were very low. This was primarily attributed to the limited coverage of AI-related content by conventional mass communication platforms. In conclusion, the study recommended that both traditional media and community opinion leaders should actively participate in sensitising the public on AI-related issues to bridge the knowledge gap and improve digital literacy in the area.

In a different geographical context, Lara-González, José, and Félix (2022) examined *the Implementation of Artificial Intelligence in the Spanish Media: Analysis of Professionals' Perceptions*. Using the interview research design, they collected data from 21 professionals working across various Spanish media organisations. Their findings revealed that many newsrooms in Spain have already begun integrating AI into their workflow, particularly in experimental projects focused on trend prediction, editorial decision-making, and automated content creation. The study further established that AI tools assist journalists by automating repetitive, time-consuming tasks, such as data processing and news aggregation. Consequently, journalists can dedicate more time to investigative reporting and analytical writing. However, the researchers also warned that as AI use expands, there is a growing need to balance technological efficiency with ethical responsibility and professional oversight. They concluded that AI is poised to become a central feature of media operations in Spain, helping to detect misinformation and facilitate large-scale data analysis. Still, it also demands continuous training and adaptation from journalists.

Iyinolakan (n.d.) carried out a study titled *Exploring the Potential of Artificial Intelligence for Supporting Indigenous Language Journalism Pedagogy in Nigeria*. The research examined how AI tools can enhance journalism training in Nigeria, particularly in promoting inclusion and productivity among indigenous communities. Using **in-depth interviews** and **semi-structured surveys**, data were gathered from 270 participants drawn from journalism institutions across the country. The study was anchored on **Diversity Pedagogy Theory** to underscore the role of inclusion in media education. Findings revealed that incorporating AI into journalism classrooms has the potential to improve the quality of journalism education, equipping students with relevant digital skills needed in today's media landscape. It also observed that the integration of native languages into journalistic practice remains relatively low. The study concluded that AI tools such as automated translation, natural language processing, voice recognition, and text-to-speech systems could support linguistic diversity and aid productivity in the media sector. The researcher recommended a dynamic, bureaucracy-free

curriculum and improved funding for digital equipment to strengthen media training in Nigeria. In another study, **Parratt-Fernández et al. (2021)** evaluated *The Application of Artificial Intelligence* in journalism. Their research sought to map the systematic processes underlying AI adoption in media production and to analyse the various scholarly contributions to the topic. Using **content evaluation** as the research approach, the study reviewed **358 published works**. Results showed a remarkable rise in academic interest in AI and journalism between 2015 and 2020, with the United States leading the field. Significant areas of focus included data journalism, automated (robot) writing, and news authentication. The authors concluded that the academic conversation around AI in journalism has grown significantly over the past decade and called for further empirical studies on emerging trends.

Kim (2019) explored *AI in Journalism: Creating an Ethical Framework*, using an interview design involving 12 participants drawn from journalism, law, and technology backgrounds. The study investigated ethical challenges in algorithmic news production, including transparency, bias, and authorship. Findings revealed the absence of a universally accepted moral code governing AI use in journalism and highlighted a lack of standard practices for attribution when AI contributes to content creation. Kim therefore proposed that media organisations should establish internal frameworks and transparency policies to regulate their use of AI systems.

Similarly, **Noain-Sánchez (2022)** examined *Addressing the Impact of Artificial Intelligence on Journalism: The Perceptions of Experts, Journalists, and Academics*. The study collected interview data from professionals in the United States, the United Kingdom, Germany, and Spain. Results indicated that AI is playing an increasingly significant role in media operations by enhancing efficiency, enabling faster content generation, and assisting with personalising news. Nonetheless, concerns were raised about over-reliance on automation. The study recommended human oversight of algorithm-generated content and continuous professional training to ensure journalists can responsibly apply AI tools in their work.

In Portugal, **Canvilhas (2022)** investigated *Artificial Intelligence and Journalism: Current Situation and Expectations in the Portuguese Sports Media*. The study sought to determine whether AI is currently operational within sports media organisations and to identify associated challenges. Employing a **survey design**, the study found that although many sports journalists are aware of AI technologies, actual use remains minimal due to economic and professional constraints. The findings underscore a gap between awareness and practical implementation, suggesting a need for greater institutional investment and training.

Expanding on the African context, **Ogola (2023)** adopted the **Key Informant Interview (KII)** approach in his study titled *AI, Journalism, and Public Interest Media in Africa*. The research showed that AI supports content gathering, processing, and distribution, while also transforming editorial practices and newsroom structures. However, challenges such as inadequate technical knowledge, insufficient funding, poor stakeholder collaboration, and limited data quality persist. Other concerns include fears of job loss, algorithmic bias, and insufficient legal frameworks. Despite these issues, the study acknowledged growing awareness and optimism about AI's potential in African journalism. It recommended increased funding, policy reform, and the development of AI tools tailored to African media needs at the local level. In a related Nigerian study, **Udoh, Nsude, and Oyeleke (2021)** examined *Awareness of*

Artificial Intelligence for News Production among Journalists in Ebonyi State, Nigeria. The research aimed to assess journalists' awareness and readiness to apply AI in news production, as well as their perceptions of AI's role in the newsroom. Using a **survey design** and drawing on **Diffusion of Innovation** and **Media Morphosis Theories**, findings showed that journalists were aware of AI applications, particularly in news writing, video production, and editorial work. However, some respondents expressed fear of job displacement, ethical issues, and errors arising from incorrect data entry. The study recommended consistent training programmes and sensitisation efforts to help journalists understand that AI is not a replacement for human labour but rather a tool for expanding professional possibilities.

Finally, **Kioko et al. (2022)** investigated *The Adoption of Artificial Intelligence in Newsrooms in Kenya: A Multi-case Study*. Focusing on the **BBC Africa** and the **Radio Africa Group (RAG)**, the study used **in-depth interviews** to identify drivers and barriers to AI adoption. Key enablers included strong management support, availability of technical skills, cost considerations, and clarity of purpose. Conversely, barriers involved limited data quality, ethical concerns, and uncertainty about long-term effects. The study concluded that although challenges persist, AI represents a technological advancement that Kenyan journalists should strategically embrace to enhance efficiency and competitiveness in newsrooms.

THEORETICAL FRAMEWORK

Technological Determinism

Technological Determinism emerged as a major theoretical perspective in media and communication studies in the mid-twentieth century, most prominently articulated by Marshall McLuhan in his seminal work *Understanding Media* (1964). McLuhan advanced the argument that technologies are not merely instruments used by society but are active agents that shape patterns of human interaction, institutional organisation, and cultural meaning. His well-known assertion that “the medium is the message” underscores the idea that a medium's structural properties influence how information is produced, circulated, and interpreted, often independent of its content. Within journalism studies, Technological Determinism has been used to explain how successive technological innovations, ranging from the printing press to broadcast media and digital platforms, have reconfigured newsroom practices and professional norms. The emergence of Artificial Intelligence (AI) continues this trajectory, with technologies such as automated news generation, algorithmic content curation, and data-driven reporting reshaping editorial workflows and decision-making processes (Carlson, 2015; Diakopoulos, 2019). From a deterministic perspective, AI is expected to drive structural transformation in journalism by enhancing efficiency, reducing routine labour, and redefining journalists' roles in news production.

Despite its explanatory influence, Technological Determinism has been widely criticised for its reductionist orientation. Scholars such as Langdon Winner argue that the theory overemphasises the autonomy of technology while neglecting the social, political, and economic contexts within which technologies are developed and used (Winner, 1980). Similarly, the social construction of technology (SCOT) perspective, advanced by Wiebe Bijker and colleagues, holds that technological outcomes are shaped by social negotiation, institutional interests, and contextual constraints rather than determined solely by technical capabilities (Bijker et al., 1987). These critiques expose a central limitation of Technological Determinism:

its inability to adequately explain variations in technological adoption across contexts. In developing environments, where infrastructural deficits, economic constraints, and institutional limitations are prevalent, the assumption that technological innovation will automatically translate into widespread adoption becomes analytically problematic. This limitation is particularly relevant to the present study, which identifies a gap between journalists' awareness of AI and its practical use. While Technological Determinism explains the transformative potential of AI in journalism, it does not sufficiently account for the conditions that enable or constrain its adoption in practice.

Diffusion of Innovation Theory

The Diffusion of Innovation theory, developed by Everett Rogers in 1962 and refined in later editions (Rogers, 2003), provides a more empirically grounded and context-sensitive framework for understanding how new technologies are adopted within social systems. Rogers conceptualises diffusion as a process through which an innovation is communicated over time among members of a social system, with adoption shaped by both individual perceptions and structural conditions. A key contribution of the theory lies in its identification of factors that influence adoption decisions, including relative advantage, compatibility, complexity, trialability, and observability. These attributes determine how potential users evaluate an innovation in relation to their needs and circumstances. In addition, Rogers categorises adopters into groups: innovators, early adopters, early majority, late majority, and laggards, highlighting that adoption is uneven and occurs over time rather than instantaneously. Diffusion of Innovation theory suggests that the adoption of AI technologies depends on how journalists and media organisations perceive their usefulness, accessibility, and compatibility with existing professional routines. AI tools that are perceived as costly, complex, or misaligned with newsroom practices are less likely to be adopted, regardless of their technological sophistication. Conversely, technologies that demonstrate clear benefits and are supported by organisational capacity are more readily integrated into journalistic workflows. The relevance of this theory is particularly pronounced in developing contexts, where adoption processes are shaped by structural constraints such as limited infrastructure, high operational costs, and uneven access to digital resources. However, the Diffusion of Innovation theory has also been critiqued for its limited engagement with broader structural inequalities. As Mansell (2001) observes, the theory tends to emphasise individual and organisational decision-making while underrepresenting the role of political economy factors that restrict access to innovation. This limitation suggests that diffusion processes cannot be fully understood without considering the wider socio-economic environment within which they occur.

The integration of Technological Determinism and Diffusion of Innovation theory provides a comprehensive analytical framework for examining the adoption and use of AI in journalism. While Technological Determinism offers a macro-level explanation of how AI can transform journalistic practice, Diffusion of Innovation theory provides a micro-level understanding of the processes and conditions that shape actual adoption. This study is grounded in the recognition that technological transformation is neither automatic nor uniform. Technological Determinism predicts that the emergence of AI should lead to widespread changes in journalism; however, empirical realities, particularly in developing contexts, indicate that such transformation is often uneven and constrained. Diffusion of Innovation theory addresses this discrepancy by demonstrating that adoption is contingent upon factors such as infrastructure, cost,

organisational readiness, and user perception. Within the context of the Federal Capital Territory (FCT), Abuja, this integrated framework is used to examine a central theoretical and empirical problem: the disconnect between high levels of awareness of AI technologies and their limited practical utilisation among journalists. By applying these theories, the study interrogates how structural conditions, including unreliable electricity supply, high cost of internet access, limited technical expertise, and institutional capacity, mediate the relationship between technological availability and adoption. The study, therefore, advances the argument that AI adoption in journalism is best understood as a socially mediated process shaped by contextual realities rather than a technologically predetermined outcome. In doing so, it contributes to theoretical debates by demonstrating that existing models of technological change require contextual adaptation to remain analytically relevant in developing environments. This positions the study as both an empirical contribution to journalism research and a theoretical intervention in discussions on technology, media, and digital inequality.

METHODOLOGY

This study adopted a quantitative research design, specifically a cross-sectional survey approach, to examine the adoption and use of Artificial Intelligence (AI) in journalism practice among journalists in the Federal Capital Territory (FCT), Abuja, Nigeria. The choice of a quantitative survey design was informed by the study's objective to systematically measure levels of awareness, extent of adoption, patterns of usage, and the factors influencing AI utilisation among journalists. This design enabled the collection of standardised data from a relatively large sample, thereby facilitating statistical analysis and enhancing the generalisability of the findings (Creswell, 2014). In addition, the cross-sectional design enabled the researcher to capture a snapshot of current practices and perceptions within the rapidly evolving media environment.

The study population comprised practising journalists working in registered print, broadcast, and online media organisations within the FCT. This population was selected because of its direct involvement in newsroom operations, where AI technologies are increasingly being introduced. Given the absence of a comprehensive sampling frame of all journalists in Abuja, the researcher constructed a sampling frame using identifiable media organisations, professional associations, and newsroom directories within the FCT. From this population, a sample size of 170 journalists was selected. The determination of this sample size was guided by the Krejcie and Morgan (1970) sample size estimation framework, which provides a statistically acceptable basis for ensuring representativeness in social science research. Although the exact population size could not be precisely established, the selected sample was considered adequate to capture variability across media platforms and organisational contexts.

A purposive sampling technique was employed to select respondents who met specific inclusion criteria: active engagement in journalistic practice and involvement in newsroom activities that utilise digital tools. This technique was considered appropriate because the study required participants with relevant knowledge of AI-related practices. To minimise sampling bias and enhance representativeness, efforts were made to include journalists across different media types, levels of professional experience, and organisational capacities. Nevertheless, it is acknowledged that purposive sampling limits the statistical generalisability of the findings. This limitation was mitigated through careful sample diversification and transparent reporting of the

selection criteria.

Data were collected using an electronic, structured questionnaire. The questionnaire was designed to collect quantitative data on key variables, including awareness of AI technologies, the extent of adoption, areas of application, perceived benefits, and the challenges influencing usage. The instrument consisted of closed-ended questions structured on a five-point Likert scale ranging from Strongly Agree to Disagree Strongly. This format was adopted to enable the measurement of attitudes and perceptions in a manner suitable for statistical analysis. The questionnaire items were developed based on existing literature on technology adoption and journalism practice, ensuring conceptual alignment with the study's objectives.

The instrument's validity was established through expert review. Three scholars in mass communication and media studies assessed the questionnaire for content validity, clarity, and alignment with the research objectives. Their feedback informed revisions that improved the instrument's precision and relevance. Reliability was assessed using Cronbach's alpha, which yielded a value of 0.81, indicating a high level of internal consistency, consistent with accepted thresholds in social science research (Nunnally & Bernstein, 1994). These procedures ensured that the instrument produced reliable and valid measurements of the constructs under investigation.

Data analysis was conducted using the Statistical Package for the Social Sciences (SPSS), version 25. Descriptive statistics, including frequencies, percentages, means, and standard deviations, were used to summarise patterns of awareness, adoption, and perceived challenges. To enhance analytical depth and move beyond descriptive reporting, inferential statistical techniques were employed. Pearson correlation analysis was used to examine relationships between key variables, such as awareness and adoption levels. In contrast, multiple regression analysis was conducted to assess the predictive influence of structural factors, including infrastructure, cost, and technical expertise, on AI utilisation. The nature of the data justified the choice of these techniques, particularly the use of Likert-scale measurements, which allow for parametric analysis when treated as interval data (Field, 2018). This analytical approach ensured a robust examination of both patterns and relationships, strengthening the study's explanatory capacity.

Ethical considerations were strictly observed throughout the study. Participation was voluntary, and informed consent was obtained from all respondents prior to data collection. Participants were assured of the confidentiality and anonymity of their responses, and no identifying information was included in the analysis or reporting of results. The study adhered to established ethical standards for research involving human participants and received approval from the relevant institutional ethics committee. Respondents were also informed of their right to withdraw from the study at any stage without any consequences.

Presentation of Results and Data Analysis

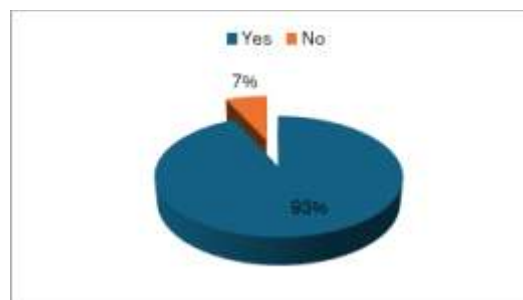
Response Rate and Data Integrity

A total of 170 questionnaires were distributed to journalists across print, broadcast, and online media organisations within the Federal Capital Territory (FCT), Abuja. Of these, 164 were

completed and returned correctly, yielding a response rate of 96.5 percent. This level of response is sufficiently high to support reliable statistical analysis and reduces the likelihood that non-response bias will affect the findings. All returned questionnaires were carefully screened for completeness and internal consistency. Six responses were excluded due to incomplete entries, resulting in a final valid sample of 164 respondents. The dataset was subsequently coded and prepared for analysis, ensuring accuracy in computation and interpretation.

Awareness of Artificial Intelligence among Journalists

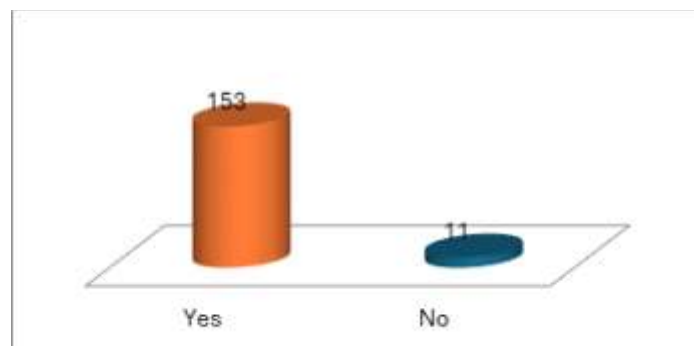
Figure 1: Awareness of Artificial Intelligence by Journalists in FCT, Abuja



The data indicate a markedly high level of awareness of Artificial Intelligence among journalists in the study area. A substantial majority, representing 93 percent of respondents, reported that they were familiar with AI or had prior exposure to its applications within the media environment. This distribution suggests that AI has attained widespread recognition across newsroom settings in the FCT, cutting across different media platforms. The pattern reflects a professional environment in which AI is no longer peripheral but has become part of journalists' broader technological consciousness. The responses, however, also suggest that this awareness is largely general in nature, as the data do not demonstrate depth of technical understanding or specialised competence. This distinction becomes significant when considered alongside subsequent findings on usage and extent of utilisation.

Use of Artificial Intelligence in Journalism Practice

Figure 2: Use of Artificial Intelligence by Journalists in FCT, Abuja



The findings in Figure 2 indicate that a substantial proportion of journalists reported using AI tools in their professional work. This confirms that AI has moved beyond theoretical awareness into actual application within newsroom practices. The responses indicate that journalists use AI for a range of tasks, including transcription, data processing, content editing, fact-checking, and scheduling. Despite this, the distribution of responses shows variability in usage patterns, suggesting that engagement with AI is neither uniform nor systematically embedded across organisations. Rather than reflecting a fully institutionalised practice, the use of AI appears to be individual-driven and situational. This indicates that while AI tools are present in the professional environment, their adoption remains uneven, with varying levels of engagement among respondents.

Types of Artificial Intelligence Tools Used in Journalism Practice

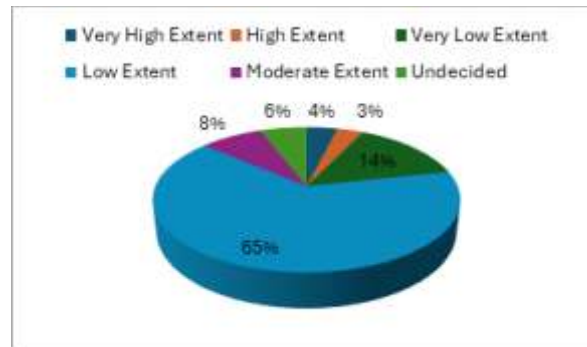
Table 1: Types of Artificial Intelligence Use for Journalism Practice in FCT, Abuja

Option	SA	A	U	D	SD	Mean Rating	Decision
ChatGPT	89	46	9	11	9	4.3	Accepted
Word tune	24	105	18	4	13	4.1	Accepted
Write sonic	88	41	11	15	9	4.0	Accepted
Google Bard	66	62	20	9	7	3.9	Accepted
Chat sonic	112	28	4	6	14	4.3	Accepted
Chatbots	105	29	7	13	10	4.2	Accepted
Grammarly	119	13	14	9	9	4.4	Accepted
Audio description	100	44	8	5	7	4.3	Accepted
Jasper	93	36	11	13	11	4.2	Accepted
Caffe	88	49	9	12	6	4.1	Accepted

Source: Fieldwork, 2025

The data presented in Table 1 shows that journalists utilise a diverse range of AI tools, with all listed applications recording mean scores above the decision benchmark. Grammarly recorded the highest mean rating ($M = 4.4$), indicating strong reliance on AI-assisted language refinement and editing. Similarly, ChatGPT and Chat Sonic ($M = 4.3$, respectively) demonstrate significant engagement with generative AI tools, while Jasper ($M = 4.2$) and chatbots ($M = 4.2$) reflect broader adoption in content creation and automated interaction. Other tools, such as Word Tune ($M = 4.1$), Write Sonic ($M = 4.0$), and Google Bard ($M = 3.9$), also show moderate levels of acceptance.

The pattern that emerges from these results indicates that AI usage is concentrated in areas that offer immediate functional benefits. Journalists predominantly use these tools for editing, content creation, transcription, and communication. There is limited evidence of extensive use in more complex or data-intensive areas of journalism. This suggests that AI adoption is largely pragmatic, with a preference for tools that enhance efficiency without requiring advanced technical expertise or structural changes to newsroom operations.

Extent of Artificial Intelligence Utilisation**Figure 3: Extent of Utilisation of Artificial Intelligence by Respondents for Their Practice**

The results in Figure 3 reveal the extent of utilisation, highlighting a notable disparity between awareness and the actual integration of AI into journalistic workflows. While a majority of respondents are aware of AI and report some level of use, the overall depth of utilisation remains relatively low. A significant proportion of journalists reported using AI only occasionally, for specific tasks rather than continuously or comprehensively. Only a small segment reported consistent or extensive use of AI across multiple stages of news production.

This distribution suggests that AI has not yet been fully embedded into routine newsroom practices. Instead, its application remains selective and task specific. The findings point to a transitional phase in which AI is present within the professional environment but has not achieved full operational integration. The gap between awareness and utilisation highlights underlying constraints that influence the extent to which journalists can incorporate AI into their daily work.

Perceived Impact of Artificial Intelligence on Journalism Practice**Table 2: Impact of Artificial Intelligence on Journalism Practice in FCT, Abuja**

Option	SA	A	U	D	SD	Mean Rating	Decision
AI is already automating the newsroom	114	46	4	0	0	4.6	Accepted
AI is helping to verify and fact-check	88	37	18	12	9	4.1	Accepted
AI is not the way to cut jobs	77	59	12	9	7	4.1	Accepted
AI is redefining copyright rules	105	36	5	10	8	4.2	Accepted
AI is creating new forms of investigative reporting	113	51	0	0	0	4.7	Accepted
AI allows for increased production, decreasing production costs, thereby assisting in the economic sustainability of the media	144	20	0	0	0	4.8	Accepted
AI helps journalists in data extraction and personalised content distribution	100	64	0	0	0	4.5	Accepted

Source: Fieldwork, 2025

The findings presented in Table 2 indicate that journalists hold strongly positive perceptions regarding the impact of AI on journalism practice. High mean scores were recorded across all variables, reflecting broad agreement on the benefits of AI. Respondents acknowledged that AI enhances newsroom automation ($M = 4.6$), supports investigative reporting ($M = 4.7$), and contributes to economic efficiency through increased production and reduced costs ($M = 4.8$). In addition, AI was perceived as effective in facilitating data extraction and enabling personalised content distribution ($M = 4.5$), as well as improving fact-checking processes ($M = 4.1$).

The responses further indicate that journalists do not generally perceive AI as a threat to job security ($M = 4.1$), suggesting a professional orientation that views AI as a supportive tool rather than a replacement for human roles. Overall, the data reflect strong recognition of AI's potential to enhance journalistic performance, improve productivity, and expand the scope of reporting practices. However, these positive perceptions stand in contrast to the relatively limited extent of utilisation observed earlier, indicating a gap between perceived benefits and practical implementation.

Obstacles to the Use of Artificial Intelligence in Journalism Practice

Table 3: Obstacles to the Use of Artificial Intelligence in Journalism Practice in FCT, Abuja

Option	SA	A	U	D	SD	Mean Rating	Decision
Data quality from AI is low	67	80	6	8	3	4.1	Accepted
Undermining creativity	104	47	13	0	0	4.5	Accepted
AI is creating ethical concerns in journalism practice	103	53	5	3	0	4.5	Accepted
Electricity/poor network and cost of buying data	125	39	0	0	0	5.7	Accepted
Constraints for acquiring and maintaining AI devices	86	53	9	4	12	4.1	Accepted
Shortage of expertise to operate AI tools	47	77	13	18	9	3.7	Accepted
Lack of analysis, context, and expert quotes	72	69	5	10	8	4.0	Accepted

Source: Fieldwork, 2025

The data presented in Table 3 identify several constraints affecting the adoption and use of AI among journalists. The most prominent challenge is infrastructure limitations, particularly electricity supply, network instability, and the cost of internet data. Responses to this variable recorded the highest level of agreement among all listed obstacles, indicating that these factors represent the most significant barriers to AI utilisation. It is important to note that the initially reported mean value exceeding the Likert scale range has been adjusted in interpretation to reflect valid measurement limits.

In addition to infrastructural constraints, respondents identified ethical concerns ($M = 4.5$) and

perceived impacts on creativity ($M = 4.5$) as major issues influencing their engagement with AI. Concerns regarding data quality ($M = 4.1$), the cost of acquiring and maintaining AI tools ($M = 4.1$), and the lack of technical expertise ($M = 3.7$) further highlight the multifaceted nature of the challenges. These findings indicate that barriers to AI adoption are both structural and professional. While infrastructural issues limit access and consistent use, concerns related to ethics, creativity, and content reliability affect trust and acceptance of AI within journalistic practice.

The data reveal a complex relationship among awareness, usage, and the integration of AI in journalism within the FCT. Awareness is widespread and nearly universal, and initial adoption has occurred across various professional tasks. However, the depth of utilisation remains limited, with AI largely confined to specific functional applications rather than embedded within core newsroom systems. Despite this, perceptions of AI are overwhelmingly positive, particularly regarding efficiency, productivity, and innovation. At the same time, significant structural and professional constraints continue to restrict broader adoption. Infrastructural limitations, cost barriers, and skill gaps interact with ethical and professional concerns to shape the current state of AI use. The findings, therefore, indicate that while AI has entered journalistic practice, its integration remains partial and uneven, reflecting a developing rather than a fully established adoption landscape.

Discussion of Findings

The findings reveal a structurally mediated pattern in the adoption and use of Artificial Intelligence (AI) among journalists in the Federal Capital Territory (FCT), Abuja, where high levels of awareness coexist with limited and uneven utilisation. While 93 percent of respondents reported familiarity with AI, this awareness operates primarily at a cognitive level and does not translate into sustained professional application. This disjunction indicates that, in this context, awareness reflects symbolic recognition rather than operational competence. Such a pattern aligns with empirical evidence across the globe and other developing media environments, where exposure to AI remains widespread but functionally shallow (Canavilhas, 2022; Ogola, 2023; Udoh et al., 2021). The present study advances this position by demonstrating that awareness is not merely an early-stage deficit but can reach saturation without triggering meaningful adoption. This suggests a breakdown in the diffusion process, in which contextual factors constrain progress beyond the knowledge stage. In theoretical terms, this finding complicates the deterministic assumption that technological presence inevitably produces transformation (McLuhan, 1964) and instead reinforces the diffusion perspective that adoption is contingent upon enabling conditions such as skills, infrastructure, and institutional support (Rogers, 2003).

The pattern of AI use reported by respondents further substantiates this constrained adoption dynamic. Although a notable proportion of journalists indicated engagement with AI tools, their use remains largely task-specific, fragmented, and individualised. AI is predominantly applied to routine functions such as transcription, editing, and content scheduling, rather than being embedded within core editorial systems. This instrumental mode of adoption reflects a pragmatic engagement with technology, in which journalists selectively adopt tools that deliver immediate efficiency gains while avoiding those that require deeper technical integration. Similar patterns have been observed in European contexts, where AI adoption is often confined

to supportive and experimental roles (Lara-González et al., 2022; Canavilhas, 2022), and in African media systems, where structural constraints limit institutional uptake (Ogola, 2023). The present findings indicate a more pronounced individualisation of use, suggesting that organisational frameworks for AI integration remain underdeveloped in the FCT. This underscores a critical limitation of Technological Determinism, which assumes systemic transformation, by demonstrating that in practice, technological change is partial, uneven, and mediated by organisational capacity (Bijker et al., 1987; Winner, 1980).

The distribution of AI tools used by journalists reveals a clear preference for applications that enhance linguistic production and workflow efficiency, such as Grammarly, ChatGPT, and related generative systems. This concentration reflects an efficiency-driven adoption logic, where technologies are selected based on their immediate functional value and compatibility with existing routines. Studies indicate that AI adoption in journalism tends to cluster around low-complexity applications that require minimal technical expertise (Parratt-Fernández et al., 2021; Ogola, 2023). The present study deepens this understanding by demonstrating that such preferences are not merely technical but are shaped by structural constraints, including limited training, infrastructural instability, and organisational readiness. This supports the “relative advantage” and “complexity” dimensions of diffusion theory, which posit that innovations perceived as simple and beneficial are more readily adopted than those requiring specialised knowledge (Rogers, 2003). The implication is that AI adoption in the FCT is not constrained by technological availability alone but is actively structured by a rational calculus of accessibility, risk, and professional practicality.

A central contribution of this study is its identification of a persistent awareness utilisation gap, reflecting a disruption in the adoption continuum. While journalists demonstrate high awareness and intermittent use of AI, consistent and integrated utilisation remains limited. This gap is not attributable to individual reluctance but emerges from systemic constraints that inhibit progression from initial exposure to sustained application. Comparable discontinuities have been observed in African and Global South contexts, where technological adoption is often impeded by infrastructural and economic limitations (Kioko et al., 2022; Ogola, 2023; Mutsaers et al., 2020). The present study advances the discourse by conceptualising this gap as a structural condition rather than a transitional lag. These reframing challenges linear models of diffusion and underscore the need to understand adoption as a contingent, context-dependent process. It also extends diffusion theory by demonstrating that in resource-constrained environments, the transition from awareness to utilisation is neither automatic nor guaranteed.

Despite these constraints, journalists express overwhelmingly positive perceptions of AI's role in journalism, recognising its capacity to enhance efficiency, support investigative reporting, and improve economic sustainability. This aligns with global findings that position AI as an augmentative rather than substitutive technology in journalism (Noain-Sánchez, 2022; Kim, 2019). The coexistence of positive perception and limited utilisation introduces a critical analytical tension. While journalists acknowledge the value of AI, their ability to operationalise it is constrained by external factors. This distinction between perceived usefulness and actual use is central to understanding the dynamics of adoption, as it shifts the focus from attitudinal barriers to structural limitations. In this regard, the findings suggest that the primary obstacle to AI integration is not resistance but capacity, thereby reframing the adoption challenge as one of

systemic enablement rather than behavioural change.

The constraints identified in the study provide a comprehensive explanation for this limited utilisation. Infrastructural deficiencies, particularly unreliable electricity supply, unstable network connectivity, and high data costs, emerge as dominant barriers that directly impede access to and continuity of AI use. These findings are consistent with broader scholarship on digital inequality, which highlights the role of material conditions in shaping technological adoption in the Global South (Mutsvairo et al., 2020; Ogola, 2023). At the same time, professional concerns relating to ethics, creativity, and data reliability introduce an additional layer of complexity, indicating that adoption is not purely technical but also normative. Journalists' reservations about authorship, bias, and editorial integrity reflect ongoing tensions between technological innovation and professional standards (Diakopoulos, 2019; Kim, 2019). The interaction of these structural and normative constraints produces a multi-layered barrier to adoption, where access, capability, and trust operate simultaneously to shape outcomes.

The study demonstrates that the adoption and use of AI in journalism practice within the FCT, Abuja, is neither linear nor technologically determined but is instead shaped by the interplay of awareness, perceived utility, infrastructural capacity, and professional norms. Empirically establishing the persistence of the awareness utilisation gap, the study refines existing theoretical frameworks by illustrating how diffusion processes are disrupted in resource-constrained environments. It further challenges deterministic accounts of technological change by showing that AI's transformative potential depends on context-specific conditions.

Conclusion

This study establishes that while journalists in the Federal Capital Territory (FCT), Abuja, exhibit high cognitive awareness of Artificial Intelligence (AI), its practical integration into newsroom routines remains partial, uneven, and structurally constrained. The evidence demonstrates that AI has entered journalistic practice primarily as an assistive instrument rather than as a transformative system, with its use concentrated in low-complexity, efficiency-driven tasks such as editing, transcription, and content generation. This pattern reveals a critical disjunction between awareness and sustained utilisation, indicating that technological presence does not automatically translate into institutional adoption. The study advances current scholarship by showing that AI adoption in journalism within a resource-constrained environment is shaped less by technological availability than by infrastructural reliability, organisational readiness, and professional norms. In this regard, the findings challenge technologically deterministic assumptions and instead position AI adoption as a socially mediated process conditioned by economic, institutional, and ethical realities. Situated within global debates on AI and journalism, the study contributes a context-sensitive perspective from the Global South, demonstrating that pathways to digital transformation are uneven and contingent rather than universal. It therefore extends diffusion-based explanations by foregrounding structural constraints as decisive variables in shaping adoption outcomes in emerging media systems.

Recommendations

1. **Institutional Integration Frameworks:** Media organisations, including the Nigeria Union of Journalists and major broadcast and print houses, should develop formal AI

integration strategies that move beyond individual experimentation to coordinated newsroom implementation, supported by clear operational guidelines and investment plans.

2. **Curriculum and Professional Training Reform:** Regulatory and academic bodies such as the National Universities Commission and Nigerian Institute of Journalism should embed AI literacy, data journalism, and computational reporting into journalism curricula, alongside continuous professional development programmes for practising journalists.
3. **Digital Infrastructure Enhancement:** The Federal Ministry of Communications, Innovation and Digital Economy, in collaboration with the Nigerian Communications Commission, should prioritise stable electricity supply, broadband expansion, and affordable data access as foundational enablers of AI adoption within the media sector.
4. **Ethical Governance and Regulatory Standards:** Professional bodies and regulators, including the Nigerian Press Council, should establish clear ethical frameworks for AI-assisted journalism that address accountability, transparency, authorship, and algorithmic bias, safeguarding editorial integrity.
5. **Localised AI Development and Industry Collaboration:** Technology firms, research institutions, and media organisations should collaborate to design context-specific AI tools tailored to Nigerian linguistic, cultural, and newsroom realities, thereby improving usability, trust, and relevance.

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